

DSA Patterns

& Subpatterns — Complete LeetCode Reference

28 Patterns

139 Subpatterns

100% LeetCode Coverage

#01

Brute Force

3 subpatterns

- Nested loops ($O(n^2)$, $O(n^3)$)
- Exhaustive enumeration (all pairs/triplets)
- Simulation

#02

Prefix Sum

4 subpatterns

- 1D Prefix Sum
- 2D Prefix Sum (matrix)
- Prefix Sum + HashMap (subarray sum = k)
- Difference Array (range update)

#03

Two Pointers

5 subpatterns

- Opposite ends (sorted array, two sum)
- Same direction (remove duplicates)
- Three pointers (3Sum, partition)
- Partition pointer (Dutch National Flag)
- Two pointers on two arrays (merge sorted)

#04

Sliding Window

4 subpatterns

- Fixed size window
- Variable size window (expand / shrink)
- Window + HashMap / Set (unique chars)
- Window + Frequency count (anagram, permutation)

#05

Fast & Slow Pointer

4 subpatterns

- Cycle detection (Floyd's)
- Find cycle entry point
- Find middle of linked list
- Happy number / sequence cycle

#06

Binary Search

5 subpatterns

- Classic binary search (sorted array)
- Left boundary search (first occurrence)
- Right boundary search (last occurrence)
- Rotated sorted array
- Search in 2D matrix

#07

Binary Search on Answer

4 subpatterns

- Minimize the maximum
- Maximize the minimum
- Feasibility check function (custom predicate)
- Capacity / allocation problems

#08

Recursion

4 subpatterns

- Linear recursion (factorial, fibonacci)
- Tree recursion (multiple calls)
- Tail recursion
- Mutual recursion

#09

Backtracking

6 subpatterns

- Subsets / Power set
- Permutations (with / without duplicates)
- Combinations (k of n)
- Constraint satisfaction (N-Queens, Sudoku)
- Path finding (maze, word search)
- Pruning strategies (cut branches early)

#10

Divide and Conquer

4 subpatterns

- Merge Sort style split
- Quick Select (k-th element)
- Binary splitting (power, matrix exponentiation)
- Closest pair / geometric problems

#11

Greedy

6 subpatterns

- Interval scheduling (sort by end)
- Activity selection
- Jump game pattern
- Greedy with sorting (custom comparator)
- Greedy + Heap (task scheduling)
- Exchange argument greedy

#12

Dynamic Programming (1D)

5 subpatterns

- Linear DP (house robber)
- DP with transitions from multiple states
- DP on strings (single string)
- DP + HashMap (memoization)
- State machine DP (stock buy/sell)

#13

Dynamic Programming (2D)

5 subpatterns

- Grid DP (unique paths, min path)
- Two-string DP (LCS, edit distance)
- DP on matrix / rectangle
- Interval DP (burst balloons, matrix chain)
- DP on bitmask (state compression)

#14

Knapsack Pattern

5 subpatterns

- 0/1 Knapsack
- Unbounded Knapsack (coin change)
- Bounded Knapsack
- Fractional Knapsack (greedy)
- Subset Sum / Partition Equal Subset

#15

Subsequence Pattern

5 subpatterns

- Longest Common Subsequence (LCS)
- Longest Increasing Subsequence (LIS)
- Longest Palindromic Subsequence
- Shortest Common Supersequence
- Count distinct subsequences

#16

Kadane's Algorithm

4 subpatterns

- Maximum subarray sum (classic)
- Maximum product subarray
- Circular array maximum subarray
- 2D Kadane (max sum rectangle)

#17

Merge Intervals

5 subpatterns

- Merge overlapping intervals
- Insert interval
- Non-overlapping intervals (remove minimum)
- Meeting rooms (conflict detection)
- Interval list intersections

#18

Monotonic Stack

5 subpatterns

- Next Greater Element
- Next Smaller Element
- Previous Greater / Smaller
- Largest rectangle in histogram
- Sum of subarray minimums / maximums

#19

Stack + Parenthesis Pattern

5 subpatterns

- Valid parentheses (matching)
- Minimum add to make valid
- Score / value of parentheses
- Expression evaluation (calculator)
- Decode string / nested structure

#20

Heap / Priority Queue

6 subpatterns

- Top-K elements (min / max heap)
- K-th largest / smallest
- Merge K sorted lists / arrays
- Sliding window maximum (deque / heap)
- Two heaps (median finder)
- Task / CPU scheduling

#21

BFS (Breadth First Search)

6 subpatterns

- Level-order tree traversal
- Shortest path in unweighted graph
- Multi-source BFS (0/1 matrix)
- BFS on grid (islands, rotting oranges)
- BFS with state (word ladder)
- Bidirectional BFS

#22

DFS (Depth First Search)

7 subpatterns

- Tree DFS (preorder / inorder / postorder)
- Graph DFS (visited set)
- DFS on grid (flood fill, islands)
- DFS + memoization (top-down DP)
- DFS for cycle detection
- DFS for connected components
- DFS for path finding / backtracking

#23

Topological Sort

4 subpatterns

- Kahn's Algorithm (BFS-based, in-degree)
- DFS-based topological sort
- Cycle detection in directed graph
- Course schedule / dependency resolution

#24

Union Find (Disjoint Set)

5 subpatterns

- Basic Union-Find (find + union)
- Union by Rank
- Path Compression
- Connected components (graph, grid)
- Dynamic connectivity / redundant connections

#25

Trie Pattern

5 subpatterns

- Insert & Search word
- Prefix search / autocomplete
- Wildcard / regex matching in Trie
- XOR Trie (maximum XOR)
- Word Search II (Trie + Backtracking)

#26

Bit Manipulation

6 subpatterns

- Basic operations (AND, OR, XOR, NOT, shifts)
- Check / set / clear / toggle a bit
- XOR tricks (single number, pairs)
- Bit counting (Brian Kernighan, popcount)
- Bitmask DP (subsets via bits)
- Power of 2 tricks & bit math

#27

Graph Shortest Path (Dijkstra)

5 subpatterns

- Classic Dijkstra (min heap)
- Bellman-Ford (negative weights)
- Floyd-Warshall (all pairs)
- 0-1 BFS (deque, binary edge weights)
- Modified Dijkstra with states (time, fuel, stops)

#28

Tree Traversal + Tree DP

7 subpatterns

- Preorder / Inorder / Postorder (recursive + iterative)
- Level order (BFS on tree)
- Tree DP - root to leaf paths
- Tree DP - subtree aggregation (diameter, max path)
- LCA (Lowest Common Ancestor)
- Binary Lifting / Euler Tour
- Serialize & Deserialize tree

■ Complete Summary

#	Pattern	Subpatterns
1	Brute Force	3
2	Prefix Sum	4
3	Two Pointers	5
4	Sliding Window	4
5	Fast & Slow Pointer	4
6	Binary Search	5
7	Binary Search on Answer	4
8	Recursion	4
9	Backtracking	6
10	Divide and Conquer	4
11	Greedy	6
12	Dynamic Programming (1D)	5
13	Dynamic Programming (2D)	5
14	Knapsack Pattern	5
15	Subsequence Pattern	5
16	Kadane's Algorithm	4
17	Merge Intervals	5
18	Monotonic Stack	5
19	Stack + Parenthesis Pattern	5
20	Heap / Priority Queue	6
21	BFS (Breadth First Search)	6
22	DFS (Depth First Search)	7
23	Topological Sort	4
24	Union Find (Disjoint Set)	5
25	Trie Pattern	5
26	Bit Manipulation	6
27	Graph Shortest Path (Dijkstra)	5
28	Tree Traversal + Tree DP	7
	TOTAL (28 Patterns)	139

Master these 143 subpatterns across 28 patterns for complete LeetCode coverage.